Distance-Based Clustering and Leakage-Aware Classification of NASA Exoplanets

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Machine Learning – A.A. 2025/2026

Abstract

This paper addresses the problem of discovering natural groupings among 6,150 confirmed exoplanets through a two-part pipeline combining distance-based clustering with a leakage-aware supervised extension. Potentially derived labels are excluded from the clustering fit and used only post-hoc. K-Means and agglomerative clustering with single and complete linkage are compared on two feature spaces. Although single linkage reaches a raw Silhouette of 0.823, it produces a severely imbalanced partition of sizes 6,145/2/2/1. The interpretation-oriented solution is K-Means ($k = 4$) on planetary and orbital features (Silhouette = 0.447; bootstrap mean = 0.412 ± 0.065), while $k = 2$ remains an essentially equivalent, more parsimonious alternative. The four-cluster partition yields interpretable exploratory groups. A post-hoc Adjusted Rand comparison indicates partial agreement with `planet_type` (0.307), but little agreement with stellar type or habitability. In the supervised extension, Macro-F1 drops from 1.000 to 0.490 after ablating mass and radius, providing concrete evidence of semantic leakage. All experiments are reproducible through six Jupyter notebooks.

1 Introduction

Motivation. Exoplanet research has expanded dramatically with space surveys, but the resulting catalogues mix raw physical measurements with derived labels that conflate observation and classification. Machine learning applied naively to these catalogues can produce misleading results: classifiers appear highly accurate because they reconstruct definitional rules rather than learning genuine predictive structure.

Problem statement. The analysis focuses on two questions. First: do confirmed exoplanets form natural groupings based on physical and orbital properties alone, without reference to any pre-existing taxonomy? Second: how much predictive signal for `planet_type` survives the removal of near-definitional features, namely planetary mass and radius?

Contributions. The main contributions are:

1. A feature audit separating physical measurements, derived labels, and observational metadata, with explicit exclusion criteria.
2. A comparison of K-Means with single- and complete-linkage agglomerative clustering across two feature spaces, evaluating Silhouette together with cluster-size diagnostics.
3. Bootstrap stability and a post-hoc Adjusted Rand comparison for clustering, together with a leakage-aware ablation study for the supervised task.
4. Bootstrap confidence intervals and corrected resampled t-tests, providing dependence-aware statistical context for the supervised comparison.

2 Related Work

Clustering of astronomical objects. Machine learning has long been used in astronomy for data-driven classification and taxonomy. Bailer-Jones et al. (1998) applied automated methods to stellar spectral classification, providing an early example of learned representations in astronomical catalogues. In the exoplanet domain, Fulton et al. (2017) revealed a gap in the radius distribution of small planets using the California-Kepler Survey, suggesting that exoplanet populations may contain physically meaningful subgroups. This motivates unsupervised investigation of structure in physical and orbital features. K-Means and agglomerative single/complete linkage are standard distance-based exploratory methods; internal validation uses Silhouette when external labels are deliberately excluded (Rousseeuw, 1987).

Supervised classification and leakage. Supervised methods applied to planetary catalogues commonly use derived categorical columns as direct targets without examining whether those categories are near-definitional from the available features. Kapoor and Narayanan (2023) systematically catalogued data leakage across 329 published ML studies, finding it led to overoptimistic results in a substantial fraction of cases. The ablation study is explicitly designed to measure and expose this risk by testing what happens when near-definitional features, namely mass and radius, are removed from the classification pipeline.

3 Methodology

3.1 Dataset and Feature Audit

The dataset used is *NASA Exoplanet Archive Intelligence*, a Kaggle dataset derived from records in the NASA Exoplanet Archive. It contains 6,150 rows and 31 columns. Columns are partitioned into three categories before modelling:

- **Physical/orbital/stellar features**: orbital period, planetary radius and mass, semi-major axis, catalogued planetary temperature `equilibrium_temp_k`, eccentricity, stellar temperature, radius, mass, age, surface gravity, and metallicity.
- **Derived or categorical labels**: `planet_type`, `star_type`, `orbital_period_cat`, `dist_category`, and `habitable_zone_flag`. These are excluded from fitting and used only post-hoc.
- **Identifiers and observational metadata**: `planet_name`, `host_star`, `discovery_method`, `disc_facility`, `ra`, `dec`, `disc_year`, `is_recent_discovery`, and `controversial_flag`.

For the supervised extension, `dist_from_earth_pc_log` is retained only as an explicit proxy for catalogue selection effects and is not interpreted as a causal predictor of planet type. The feature `star_vmag` is excluded because apparent visual magnitude is non-intrinsic and strongly correlated with distance (Pearson $r = 0.69$), making it a redundant observational proxy.

3.2 Preprocessing

Five variables with heavy right skew are transformed via `log1p`: `orbital_period_days`, `planet_radius_earth`, `planet_mass_earth`, `semi_major_axis_au`, and `dist_from_earth_pc`. Missing values are imputed with the median, which is robust to astrophysical outliers. All features are standardised with `StandardScaler`, required for distance-based methods.

For the supervised extension, `SimpleImputer` and `StandardScaler` are fitted exclusively on the training fold inside each `GridSearchCV` iteration with 5-fold `StratifiedKFold`, preventing test-set information from contaminating preprocessing statistics.

3.3 Clustering

Two feature sets are compared:

- **Set A**: catalogued planetary temperature, eccentricity, period (log), radius (log), mass (log), and semi-major axis (log).
- **Set B**: Set A plus six stellar properties: temperature, radius, mass, age, surface gravity, and metallicity.

K-Means with `n_init = 20` and `random_state = 42`, and agglomerative clustering with single and complete linkage, are run for $k = 2, \ldots, 8$. Silhouette is interpreted jointly with minimum cluster size, largest-cluster fraction, and singleton count. The cluster-balance diagnostic flags extreme imbalance when the smallest cluster is below max(30, 1 percent of the data) or the largest exceeds 95 percent. This is reported as a heuristic aid, not as a mathematical validity criterion. All values of k from 2 to 8 are evaluated; since $k = 2$ and $k = 4$ yield essentially equivalent Silhouette scores, the four-cluster solution is retained as an interpretation-oriented exploratory partition rather than as a statistically superior one. Bootstrap stability is assessed over 20 resamples, each using 80 percent of the data with replacement.

External labels remain excluded from model fitting and selection. They are compared with the final partition only post-hoc through the Adjusted Rand Index, which corrects pairwise partition agreement for chance (Hubert and Arabie, 1985).

3.4 Supervised Extension

The target is `planet_type`, with six classes after excluding `Unknown`, leaving 6,100 samples. Logistic Regression, Decision Tree, Random Forest, SVM with RBF kernel, and KNN are compared using an 80/20 stratified train/test split. Hyperparameters are tuned via `GridSearchCV` with 5-fold `StratifiedKFold`, optimising Macro-F1.

The final supervised feature set contains 16 features: 6 planetary/orbital, 6 stellar, and 4 contextual features: `n_stars`, `n_planets`, `multi_planet_system`, and `dist_from_earth_pc_log`. Two setups are evaluated:

- **Setup A (Full)**: all 16 features.
- **Setup B (Ablation)**: removes `planet_radius_earth_log` and `planet_mass_earth_log`.

Uncertainty is quantified via bootstrap confidence intervals, using 1,000 resamples of the test set and a 95 percent percentile CI on Macro-F1. Pairwise differences are assessed through a corrected resampled t-test on 10 paired Macro-F1 values from repeated stratified 5-fold CV with two repetitions. The selected hyperparameter configurations are cloned and refitted in every split.

4 Experiments and Results

4.1 Clustering Results

Table 1: Main clustering configurations and diagnostics.

Feature set	Method	k	Silhouette	Cluster sizes	Status
B – Planetary/Orbital/Stellar	Single	4	0.8228	6145/2/2/1	Severely imbalanced
B – Planetary/Orbital/Stellar	Complete	2	0.7667	6136/14	Severely imbalanced
A – Planetary/Orbital	**K-Means**	**4**	**0.4471**	**4054/1174/850/72**	**Selected**
A – Planetary/Orbital	K-Means	2	0.4460	4196/1954	Binary comparison

The highest raw Silhouette is not the most useful solution: single and complete linkage obtain large scores by isolating tiny groups of outliers. Among K-Means configurations, $k = 2$ and $k = 4$ on Set A achieve essentially equivalent Silhouette scores (0.4460 vs. 0.4471, difference = 0.0011). We retain $k = 4$ as an interpretation-oriented solution because it reveals additional interpretable substructure, separating Hot Jupiters from Eccentric Long-Period Giants, while $k = 2$ remains the more parsimonious alternative. In the evaluated non-degenerate K-Means configurations, Set A obtains better Silhouette scores than Set B, suggesting that the added stellar features do not improve this distance-based partition. Bootstrap stability gives mean Silhouette **0.412** and standard deviation **0.065** over 20 resamples.

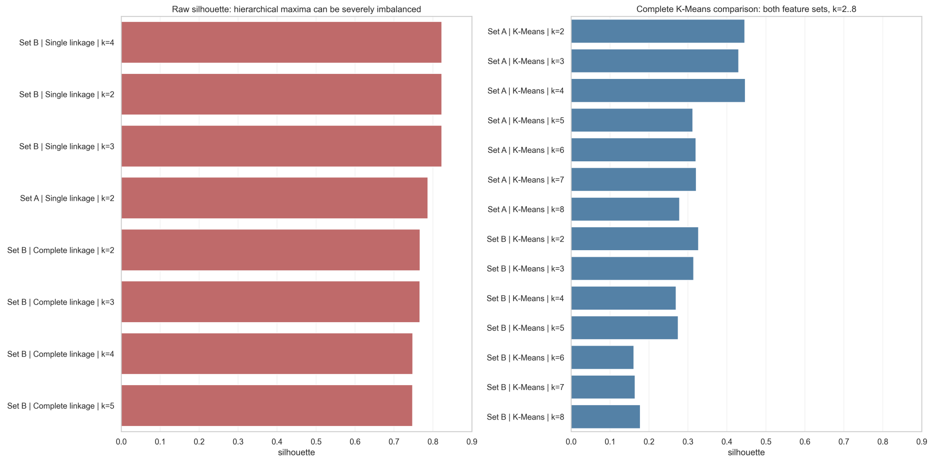

Figure 1: Silhouette scores for all method, feature set, and k combinations from $k = 2$ to $k = 8$. Single and complete linkage produce severely imbalanced partitions despite high raw scores. The selected interpretation-oriented solution is K-Means $k = 4$ on Set A, while $k = 2$ remains the more parsimonious alternative.

4.2 Cluster Profiles

Table 2: Post-hoc cluster profiles for the selected K-Means solution.

Cluster	Name	Dominant type	n	%	Temp.	Ecc.	Mass	SMA
0	Warm Mini-Neptunes	Mini-Neptune	4054	65.9	807	0.026	1.91	0.136
1	Hot Jupiters	Gas Giant	1174	19.1	1227	0.053	5.64	0.257
2	Wide-Orbit Companions	Gas Giant	72	1.2	1378	0.048	8.14	**5.773**
3	Eccentric Long-Period Giants	Gas Giant	850	13.8	771	0.286	6.59	1.139

In Table 2, cluster 2 is abbreviated as Wide-Orbit Companions for readability; the full interpretation is Ultra-Massive Wide-Orbit Companions, justified by its high log-mass and extreme semi-major axis. SMA is semi-major axis after `log1p` transformation. The bold SMA value for cluster 2 justifies the Wide-Orbit label, with a log-mean corresponding to approximately 320 AU. Cluster 3 is named Long-Period based on its orbital-period log-mean of 6.759 in `log1p` space, corresponding to approximately 861 days, the highest among the four clusters.

Cluster names are post-hoc and based entirely on mean feature profiles; no label is used in the fit. Cluster 2 is physically notable on two counts. First, its mean log-mass of 8.14 corresponds to approximately 11 Jupiter masses, near the $13 M_{\mathrm{Jup}}$ brown-dwarf boundary. Second, 69 of its 72 members were discovered by direct imaging, with median SMA approximately 325 AU and range 22–19,000 AU. This is a population whose orbital periods are typically unmeasurable and were therefore median-imputed during preprocessing. The mean period in the cluster profile reflects this imputation rather than the physical orbital timescale. Similarly, the catalogued temperature of 1,378 K is partly affected by imputation (28/72 missing) and, for self-luminous direct-imaging companions, likely reflects effective rather than irradiative equilibrium temperature; both quantities are therefore approximate descriptors. In a post-hoc sensitivity analysis, all 72 objects remain in the matched extreme group when period and temperature are jointly removed (matched group size = 81, Jaccard = 0.889; whole-partition ARI = 0.834), indicating that the group is not created solely by those two problematic features.

The external comparison is performed only after clustering. Since `planet_type` and `orbital_period_cat` are partly derived from variables included in the clustering feature space, ARI is interpreted descriptively rather than as independent ground truth. The near-zero and negative values for `star_type` and `habitable_zone_flag` indicate that the partition is not simply reproducing stellar taxonomy or habitability.

Table 3: Post-hoc Adjusted Rand comparison with external labels.

External label	Samples used	ARI	Interpretation
`planet_type` without `Unknown`	6100	0.307	Partial agreement with the dataset taxonomy
`orbital_period_cat`	6150	0.224	Partial agreement with orbital-scale categories
`star_type`	6150	0.043	Near-random agreement
`habitable_zone_flag`	6150	-0.044	Below-chance agreement

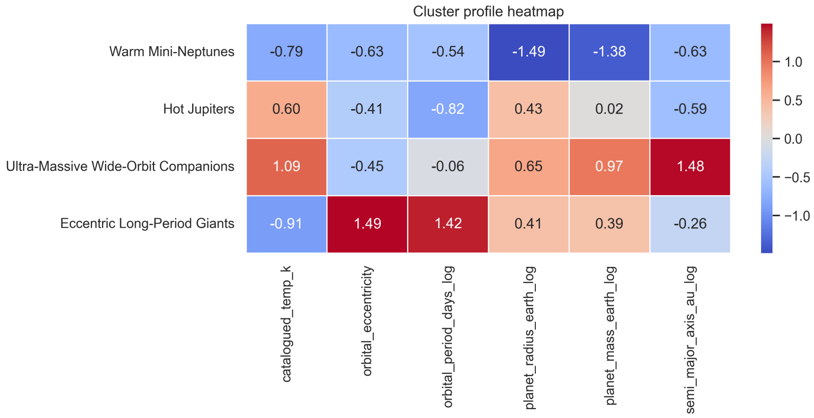

Figure 2: Z-score heatmap of mean feature profiles. Rows are named clusters; columns are the six features of Set A.

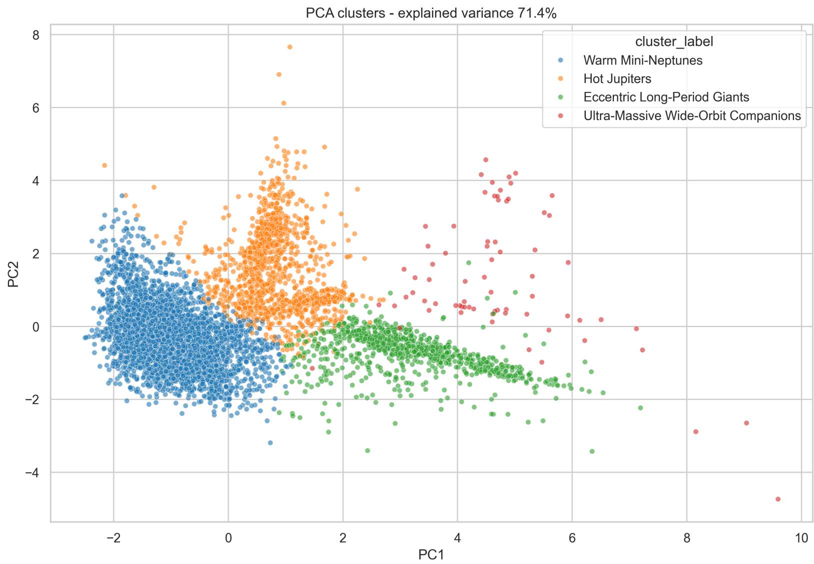

Figure 3: 2D PCA projection coloured by final cluster assignment. PCA is used only for visualisation, not as the training space.

4.3 Supervised Results

Table 4: Supervised classification results before and after leakage-aware ablation.

Setup	Model	Accuracy	Macro-F1
A – Full	Decision Tree	1.0000	1.0000
A – Full	Random Forest	0.9992	0.9989
A – Full	Logistic Regression	0.9459	0.9208
A – Full	SVM	0.9344	0.9046
A – Full	KNN	0.7746	0.6604
B – Ablation	**Random Forest**	**0.5959**	**0.4904**
B – Ablation	KNN	0.5836	0.4548
B – Ablation	SVM	0.5172	0.4484
B – Ablation	Decision Tree	0.4984	0.4315
B – Ablation	Logistic Regression	0.4779	0.4270

Setup A's near-perfect performance is not a predictive success: it demonstrates that `planet_type` is reconstructible from mass and radius, confirming semantic leakage. After ablation, Random Forest's Macro-F1 drops from 0.999 to 0.490 ($\Delta = 0.509$), the principal quantitative finding of the supervised section. KNN's weaker performance even in Setup A is consistent with its sensitivity to dimensionality and class imbalance: unlike decision-tree-based methods, KNN cannot exploit near-deterministic thresholds on mass and radius. In Setup B, the largest confusion is between Super-Earth and Mini-Neptune, with 86 and 80 errors in opposite directions, followed by Neptune-like planets predicted as Mini-Neptune in 50 cases. This is coherent with removing the mass and radius information that separates neighbouring classes.

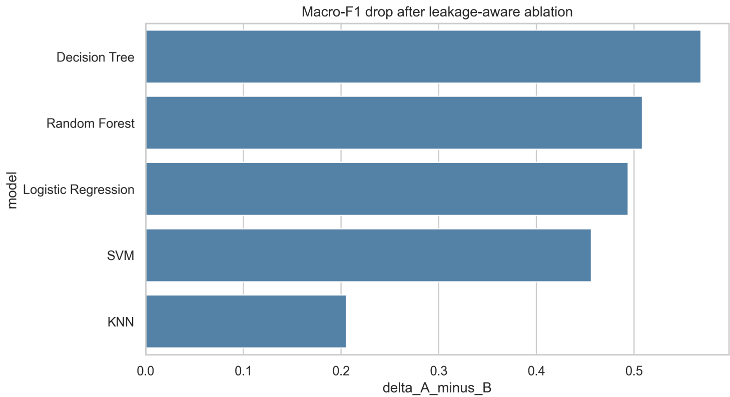

Figure 4: Macro-F1 drop from Setup A (Full) to Setup B (Ablation) for all five classifiers.

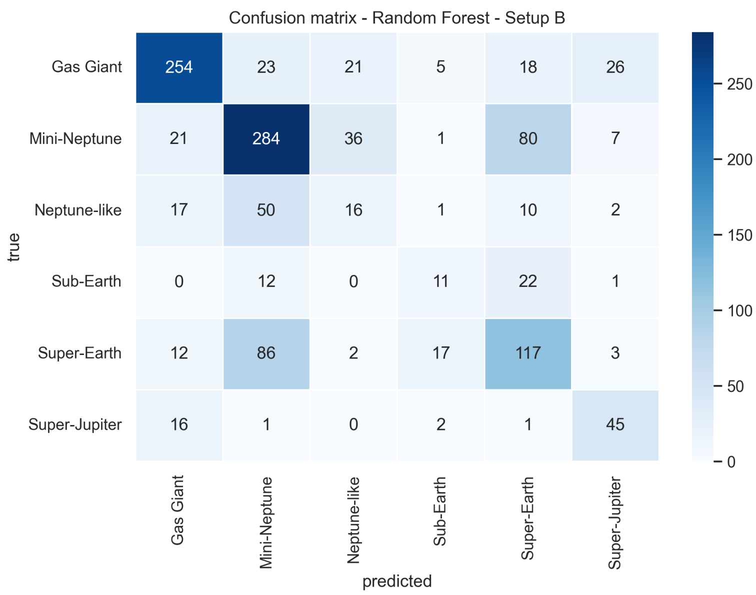

Figure 5: Confusion matrix of the best Setup B model, Random Forest. The largest errors occur between neighbouring planet classes after removing mass and radius.

4.4 Confidence Intervals and Corrected Resampled t-tests

Table 5: Bootstrap 95 percent confidence intervals on Macro-F1 for Setup B.

Model	Macro-F1	95 percent CI
Random Forest	0.490	[0.457, 0.523]
KNN	0.455	[0.422, 0.491]
SVM	0.448	[0.420, 0.479]
Decision Tree	0.432	[0.401, 0.460]
Logistic Regression	0.427	[0.399, 0.455]

Repeated stratified CV yields mean Macro-F1 values of 0.520 for Random Forest, 0.483 for KNN, 0.481 for SVM, 0.446 for Decision Tree, and 0.426 for Logistic Regression. A focused comparison shows that Random Forest exceeds KNN by 0.0368 Macro-F1 ($t = 6.754$, $p = 0.000083$). As a descriptive comparison, SVM and KNN are not significantly different ($p = 0.826$), suggesting comparable difficulty on the same instances. All comparisons are exploratory and conditional on the hyperparameter configurations selected by the primary `GridSearchCV`; the bootstrap CI and repeated-CV estimates measure different sources of uncertainty and should not be expected to coincide.

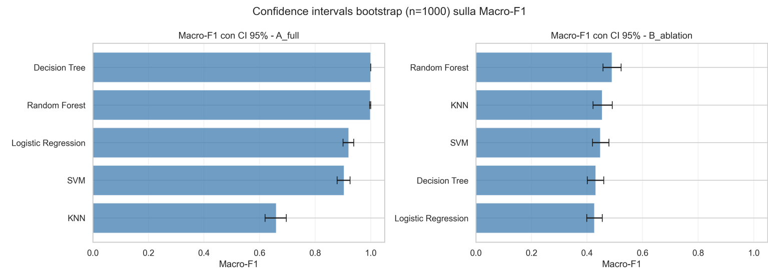

Figure 6: Bootstrap 95 percent confidence intervals on Macro-F1 for all models in both setups.

5 Discussion and Limitations

Interpretation. The clustering result indicates moderate and interpretable structure in planetary and orbital space. The silhouette of 0.447 is consistent with a continuous domain rather than sharply separated astronomical classes. As an exploratory indication, Random Forest feature importance in Setup B assigns the largest contributions to the catalogued temperature feature (0.142), orbital period (0.138), and semi-major axis (0.130). These importances are predictive, not causal; in particular, the semantics of the catalogued temperature are heterogeneous for directly imaged companions.

Limitations. First, the bootstrap analysis measures clustering-quality stability but not per-object assignment stability; the reported external-label ARI does not replace an assignment-stability analysis across matched resamples. Second, the supervised evaluation uses one stratified holdout with internal `GridSearchCV`; nested CV would reduce optimistic bias. Third, corrected t-tests use ten overlapping splits and fixed selected configurations; their unadjusted pairwise p-values are exploratory rather than confirmatory. Fourth, `planet_type` is likely constructed from physical thresholds and may not match a single published taxonomy. Fifth, the Kaggle dataset is derived from, rather than directly queried from, the NASA Exoplanet Archive; substantial missingness and the heterogeneous meaning of the catalogued temperature limit physical interpretation, and future source updates may alter numerical reproducibility.

6 Conclusions

This work presents a reproducible pipeline for unsupervised exploration and leakage-aware supervised analysis of exoplanet groups. Joint inspection of Silhouette and cluster sizes prevents severely imbalanced hierarchical partitions from being mistaken for useful structure; interpretation-oriented K-Means with $k = 4$ yields four readable groups while $k = 2$ remains a parsimonious alternative. Post-hoc ARI indicates partial, rather than redundant, agreement with existing taxonomies. The ablation study shows that near-perfect `planet_type` classification is driven by feature-label dependence. Bootstrap intervals and corrected resampled t-tests add dependence-aware statistical context. The central contribution is a methodologically defensible formulation rather than maximum performance.

Reproducibility. All code is available as six sequential Jupyter notebooks (`01_Audit` to `06_Report`) included in the submission. The accompanying notebooks contain the full hyperparameter grids, fitted preprocessing pipelines, intermediate tables, and figure-generation code; the report therefore presents only the compact experimental summary. The notebooks are executable sequentially on a standard CPU environment with no specialised hardware requirements, and the Python 3.11 dependencies are pinned in `requirements.txt`.

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